

MGTA 403 AI-Math and Programming for Business Analytics

Summer 2026

CLASS SESSIONS

- Section IDs: 278382, 278383
- Class Sessions: 7/27, 7/29, 7/31, 8/3, 8/5
- Class Session Location: 1E107, 9am - 12pm (yellow) and 1pm - 4pm (blue)

LAB SESSIONS

- Skills test: 7/22 (1N108, 1pm - 4pm)
- Practice Sessions: 7/23, 7/24 (1N108, 1pm - 4pm)
- Lab Sessions: 7/28, 7/30, 8/4, 8/6, 8/7 (1N108, 1pm - 4pm)

FINAL EXAM

- Final Exam (written and oral): 8/10 in 1E106, 1E107, 1N108 9:00a - 5:00p

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Office Hours: By appointment and during lab and practice sessions

Teaching Assistants

- Shankar Ddeenadayalan (sdeenadayalan@ucsd.edu)
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- Office hours: Available at all scheduled lab and practice sessions

DESCRIPTION

This course is intended as a gateway to mastering analytics and analytical thinking in a rapidly developing Generative AI era. Our goal is to help you build the confidence, mindset, and skills needed to tackle complex business problems using data, mathematics, programming, and Generative AI.

Unlike traditional math or coding classes, this course doesn't focus on memorizing formulas or syntax. Instead, we use Python, without any external libraries, to bring core math and statistics concepts to life. You'll learn how analytics works under the hood, and why that matters for business decision-making.

OBJECTIVES

In this course ...

- You'll code to learn: Every concept, from mean and variance to A/B testing and clustering, is something you'll implement yourself.
- You'll learn to think like an Agentic Engineer: We'll teach you to break down problems, write test cases, and build business solutions using a test-driven development (TDD) mindset.
- You'll collaborate with AI: Rather than using AI for cognitive offloading, you'll learn how to guide AI effectively as a thought partner.
- You'll solve real problems: Every concept is tied to a business context: Marketing analytics, customer segmentation, inventory decisions, and more.

By the end of this course, you won't just be better at coding or statistics; you'll be better at thinking clearly, solving problems creatively, and working with AI to build solutions that matter in the business world. Let's start building!

MATERIALS

Readings, lecture notes, and other materials will be provided through a course website. There are no required books for the class.

SCHEDULE

Date	Topic & Activities	Location	Assignments
7/22	Skills test	1N108, 1pm – 4pm	• Pre-Course Assessment
7/23	Practice Session	1N108, 1pm – 4pm	
7/24	Practice Session	1N108, 1pm – 4pm	
7/27	Class Session 1	1E107	• Individual Assignment: Programming case 0 (Part A)
7/28	Lab Session	1N108, 1pm – 4pm	
7/29	Class Session 2	1E107	
7/30	Lab Session	1N108, 1pm – 4pm	
7/31	Class Session 3	1E107	• Individual Assignment: Programming cases 1,2, and 3 (Part A) • Individual Assignment: Complete GitHub extension for case 1 (Part A)
8/3	Class Session 4	1E107	• Individual Assignment: Probability Problems 1
8/4	Lab Session	1N108, 1pm – 4pm	•
8/5	Class Session 5	1E107	• Individual Assignment: Probability Problems 2 • Individual Assignment: Programming cases 1,2, and 3 (Part B)
8/6	Lab Session	1N108, 1pm – 4pm	•
8/7	Lab Session	1N108, 1pm – 4pm	•
8/10	Final Exam (written and oral)	1E106, 1E107, 1N108 9:00am – 5:00pm	• Individual Assignment: Probability Problems 2 • Individual Assignment: Programming GitHub Extensions for cases 1,2, and 3 (Part B) • Individual Assignment: Final Exam

GRADING

The class will use a **pass/fail (i.e., S/U)** grading scheme with a final exam. Assignments due during the course will focus on learning rather than grading performance. The goal is improvement and, ultimately, mastery of key concepts.

Some of your assignments and exercises will not place any restrictions on the use of the internet and GenAI. Others, however, will. Each assignment will state explicitly what tools can and cannot be used. All assignments will be individual, so all submissions should be your own work.

Class participation will be an important component of the class. Attending class and lab sessions will contribute to class participation, but class participation is (much) more than being present in a room. The rubric that will be used to evaluate class participation is shown below:

Class Etiquette and Punctuality

Computers may not be used in class unless we are working on a quiz or exercise that requires your laptop. It is distracting to classmates to sit next to or behind someone while they browse the web, send emails, etc. However, tablets may be used to take notes. Phones can be used only to answer iClicker questions.

Class will start on time. It is distracting to classmates and to the instructor when someone arrives late, has to get to their seat, and settle in while others are trying to engage with the class material. If you are absent or late more than once, your class participation grade will be affected.

Attendance

Learning to articulate your understanding of business analytics concepts and to evaluate and respond to the work conducted by others is an important part of this class. If you miss class, you will miss this learning opportunity, and there is not an easy way to “make it up.” As a result, you should make every effort to attend each class and lab session. If you miss class or are late more than once, it will lower your class participation grade. Note that participating in Lab Sessions will also count towards your participation score in this class.

Note that UCSD provides exceptions for religious holidays, funeral attendance, and student or dependent hospitalization. If you must miss a class or lab session, please let us know through a (private) post on the Q&A page. You should still prepare and turn in assignments on time. You should also review all class and lab materials and view the class recording linked through Canvas (see the Panopto tab).

Lab Sessions

We will have several lab (and practice) sessions. You should think of these as opportunities to practice and refine the skills that you will rely on heavily throughout the MSBA program. Attendance and participation will count towards your overall class participation score.

Prep Work

MGTA 403 will build on the prep work listed below. If there were any topics that you struggled with, it would be a good idea to review the material again before we start the class.

DataCamp

You will have free access to DataCamp for the duration of the Rady MSBA program. Additional information about the login process and the assignments to be completed before the start of summer classes were shared through the Rady-Ready Canvas site. The free signup link is also provided below:

https://www.datacamp.com/groups/shared_links/9717f47706409588024fa1b84d7321f18d4313a72ddc1fb62157f8f6a6c5f9baLinks to an external site.

Note: This class requires completion of **all** assigned work on DataCamp

Math and Stats Bootcamp

All incoming MSBA students must complete a self-paced, online math & stats boot camp to ensure everyone is adequately prepared to handle the rigor of the MSBA program. The boot camp is an essential review of basic mathematical and statistical concepts relevant in a business context.

To log in to the site linked below, please use the first part of your @ucsd.edu email address as the username and your AD/email password (e.g., if your email address is aaa111@ucsd.edu, use aaa111 as your username).

<https://rsm-django-02.ucsd.edu/msba-bootcamp/Links to an external site.>

Probability and Statistics Prep

An excellent resource to refresh and expand your knowledge of Probability and Statistics is the “OpenIntro Statistics” textbook. It is in its 4th edition and authored by Diez, Çetinkaya-Rundel, and Barr. You can download it from this website: <https://leanpub.com/openintro-statistics>Links to an external site. Note that the book is offered in a pay-what-you-want format so you can get it for as low as \$0.

We recommend you review Chapters 1-5. This will ensure that you understand data basics, probability distributions, random variables, and the foundations of statistical inference. If you have time, you can work through Chapters 6-9 which cover various statistical tests and regression. Keep in mind that MGTA 453: Business

Analytics will cover statistical tests and regression but also dynamic decision making, simulation, optimization, and so on. Therefore, you may find it helpful to give Chapters 6-9 some attention during your preparation for fall.

Final Exam

Like the assignments and exercises, some elements of the final exam will allow the use of GenAI tools but others will not. You should aim for a level of mastery that will allow you to do the work both with and without AI support. The final exam will have both a written and an oral component. Details will be shared during class sessions.

Generative Artificial Intelligence (AI) policy

You will use Generative AI tools in this class. Learning to leverage Generative AI in your work is critical. These tools have considerable potential to streamline and augment our work and you should use them with the following caveats and conditions:

- GenAI tools should enhance your work, not replace it. For instance, large language models can improve the quality of your writing, identify new sources, provide high-level summaries of complex ideas or terms, generate code, etc. These tools can help you come up with new ideas, overcome writer's block, or refine your original ideas. However, you should not use GenAI tools to generate content that you simply put your name on. This constitutes an academic integrity violation, much like taking another individual's work and misrepresenting it as your own.
- We expect all students to use GenAI for class preparation, help with assignments, etc. However, we require that you fully understand and can explain, in-person or on video, all details of every line of code and every mathematical calculation that you submit as your own work.
- Large language models have documented issues with hallucination and inaccuracy (e.g., generating fake Python packages, failing logical arguments, misrepresenting sources, etc.). You are fully responsible for all content you submit in your assignments and will be graded accordingly.

How can you use tools like Claude Code to enhance your learning experience in this class?

Coding: Generative AI tools like Claude Code are already quite good at coding and are rapidly improving. "[Vibe Coding](#)[Links to an external site.](#)" is a viable option to generate code but getting the most out of these tools requires clear guidance and a solid understanding of how to build, structure, and test code. A better approach to build a solid code-base is "[Agentic Engineering](#)[Links to an external site.](#)".

Conceptual Understanding: Using GenAI and Machine Learning requires an in-depth understanding of many business and technical concepts. Generative AI tools can be incredibly useful to develop a deeper understanding of these concepts. All you need to get started is to strike up a conversation with the chatbot about topics we cover in class.

For example, start a chat in the Claude app on your phone by clicking the icon on the bottom right. **A great topic for a first discussion is Test Driven Development (TDD) in Python**

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.